

Simple Linear Regression and Correlation

MS1411: Mathematical Statistics

Based on notes by Carles Quilis Alfonso & Walpole et al.

Blekinge Institute of Technology

Spring 2025 (Updated)

References and Reading

Primary Reference

Walpole et al., "Probability & Statistics for Engineers & Scientists," 9th ed., Pearson.

Suggested Reading for this Lecture

Sections 11.1-4 (Regression Basics, Least Squares)

Sections 11.10-12 (Correlation, R-squared, Transformations)

Lecture Objectives

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- 6 **Transformation:** Recognize when a straight line isn't appropriate and how transforming data can help.

Introduction - What is Simple Linear Regression?

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To study and model the relationship between two *quantitative* (numerical) variables. We want to see if one variable can help us predict or explain the other.

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- **Dependent Variable (Y):** Also called response variable. This is the variable we want to predict or explain. It's often treated as random.

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The "Simple Linear" Part

- **"Simple":** We only use *one* independent variable (x) to predict Y . (Multiple regression uses more than one x).
- **"Linear":** We assume the average relationship between x and Y can be reasonably approximated by a *straight line*.

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- We expect a positive relationship, possibly linear.
- However, even students studying the same amount (x) will likely have different scores (Y) due to other factors (prior knowledge, test anxiety, luck, etc.). This is where the randomness in Y comes from.
- **Regression Curve/Line:** Represents the *average* exam score we'd expect for any given number of study hours.

The Simple Linear Regression Model (Population)

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$$Y = \beta_0 + \beta_1 x + \epsilon$$

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- ϵ : The **Random Error Term**. Represents the difference between the observed Y and the average Y predicted by the linear part ($\beta_0 + \beta_1 x$). It captures all other factors affecting Y besides x .

Note: β_0 and β_1 are unknown population parameters we want to estimate.

Error Term (ϵ) Assumptions (Part 1)

Why Assumptions? ** Needed for valid statistical inference (tests, CIs).

- 1 **Mean Zero:** $E(\epsilon_i) = 0$ for all i .
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- ❷ **Constant Variance (Homoscedasticity):** $Var(\epsilon_i) = \sigma^2$ for all i .
 - The spread (variance) of the errors is the same regardless of the value of x .

Error Term (ϵ) Assumptions (Part 2)

 **Uncorrelated Errors:** $Cov(\epsilon_i, \epsilon_j) = 0$ for $i \neq j$.

- The error for one observation doesn't provide information about the error for another observation. (Often relevant for time series data).

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- $\epsilon_i \sim N(0, \sigma^2)$. Important for constructing t-tests and F-tests. (Less critical with large sample sizes due to Central Limit Theorem).

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We check these assumptions using residual plots later!

Population vs. Estimated Line

Population Line (Theoretical)

$$\mu_{Y|x} = E(Y|x) = \beta_0 + \beta_1 x$$

- Describes the *true average* relationship.
- β_0, β_1 are unknown constants (parameters).
- We can never know this line perfectly unless we measure the entire population.

Estimated Line (Sample)

$$\hat{y} = b_0 + b_1 x$$

- Calculated from our sample data $(x_1, y_1), \dots, (x_n, y_n)$.
- b_0 is our estimate for β_0 .
- b_1 is our estimate for β_1 .
- $\hat{y}$ is the *predicted* value of Y for a given x , based on our sample line.
- This is the line we actually calculate and use.

Residuals: Measuring Prediction Errors

Definition

The residual (e_i) for the i -th data point is the difference between the *observed* y_i and the *predicted* $\hat{y}_i$ from our estimated line.

$$e_i = y_i - \hat{y}_i = y_i - (b_0 + b_1 x_i)$$

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Interpretation

- Residuals are the sample counterparts of the unknown errors (ϵ_i).
- They represent the "leftover" variation in y_i after accounting for the linear relationship with x_i .
- Geometrically: The vertical distance from the data point (x_i, y_i) to the fitted regression line.
- Small residuals indicate the line fits the data points well. Large residuals indicate a poor fit for those points.

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Least Squares Estimation - The "Best Fit" Idea

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The Least Squares Criterion

We define "best" as the line that minimizes the sum of the *squared* residuals (prediction errors). We square the errors so that positive and negative errors don't cancel out, and larger errors are penalized more heavily.

$$\text{Minimize: } SSE = \sum_{i=1}^n e_i^2 = \sum_{i=1}^n (y_i - \hat{y}_i)^2 = \sum_{i=1}^n (y_i - b_0 - b_1 x_i)^2$$

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The Method

Calculus is used (taking partial derivatives of SSE with respect to b_0 and b_1 , setting them to zero) to find the specific values of b_0 and b_1 that achieve this minimum SSE. These are called the least squares estimators.

Least Squares - Calculation Components

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- **Sum of Squares for x (S_{xx})**: Measures the total variability in the x values.

$$S_{xx} = \sum_{i=1}^n (x_i - \bar{x})^2 = \sum_{i=1}^n x_i^2 - \frac{(\sum_{i=1}^n x_i)^2}{n} \quad (\text{Computational Formula})$$

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- **Sum of Cross-products (S_{xy}):** Measures how x and y vary *together* relative to their means. Sign indicates direction of association.

$$S_{xy} = \sum_{i=1}^n (x_i - \bar{x})(y_i - \bar{y}) = \sum_{i=1}^n x_i y_i - \frac{(\sum_{i=1}^n x_i)(\sum_{i=1}^n y_i)}{n} \quad (\text{Computational Formula})$$

Note: Computational formulas are usually easier for hand calculations.

Least Squares - Estimator Formulas

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The estimated slope is the ratio of the co-variability (S_{xy}) to the variability in x (S_{xx}).

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Sum of Squared Errors (SSE) Calculation

Once b_1 is found, SSE (the minimized sum of squared residuals) can be calculated efficiently:

$$SSE = S_{yy} - b_1 S_{xy}$$

This represents the amount of variability in Y *not* explained by the linear relationship with x . Smaller SSE means better fit.

New Example: Advertising Spend vs. Sales

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1	2	5
2	3	7
3	1	3
4	5	11
5	4	9

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Goal: Find the least squares regression line $\hat{y} = b_0 + b_1x$.**

Example: Advertising vs. Sales (Calculations - Step 1: Sums)

Calculation Table:**

x	y	x^2	y^2	xy
2	5	4	25	10
3	7	9	49	21
1	3	1	9	3
5	11	25	121	55
4	9	16	81	36
$\sum x = 15$	$\sum y = 35$	$\sum x^2 = 55$	$\sum y^2 = 285$	$\sum xy = 125$

$$n = 5$$

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$n = 5$ Means:**

$$\bar{x} = \frac{\sum x}{n} = \frac{15}{5} = 3$$

$$\bar{y} = \frac{\sum y}{n} = \frac{35}{5} = 7$$

Example: Advertising vs. Sales (Calculations - Step 2: S_{xx} , S_{yy} , S_{xy})

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$$S_{xx} = \sum x^2 - \frac{(\sum x)^2}{n} = 55 - \frac{(15)^2}{5} = 55 - \frac{225}{5} = 55 - 45 = 10$$

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- S_{yy} (**SST**):

$$S_{yy} = \sum y^2 - \frac{(\sum y)^2}{n} = 285 - \frac{(35)^2}{5} = 285 - \frac{1225}{5} = 285 - 245 = 40$$

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- S_{xy} :

$$S_{xy} = \sum xy - \frac{(\sum x)(\sum y)}{n} = 125 - \frac{(15)(35)}{5} = 125 - \frac{525}{5} = 125 - 105 = 20$$

Example: Advertising vs. Sales (Calculations - Step 3: b_1 , b_0 , Line)

Using $S_{xx} = 10$, $S_{yy} = 40$, $S_{xy} = 20$, $\bar{x} = 3$, $\bar{y} = 7$:

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Using $S_{xx} = 10$, $S_{yy} = 40$, $S_{xy} = 20$, $\bar{x} = 3$, $\bar{y} = 7$:

- **Slope (b_1):**

$$b_1 = \frac{S_{xy}}{S_{xx}} = \frac{20}{10} = 2$$

Interpretation: For each additional €100 spent on advertising, sales are predicted to increase by €2000 (or 2 thousand Euros), on average.

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- **Intercept (b_0):**

$$b_0 = \bar{y} - b_1\bar{x} = 7 - (2)(3) = 7 - 6 = 1$$

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- **Estimated Regression Line:**

$$\hat{y} = b_0 + b_1x \implies \hat{y} = 1 + 2x$$

Example: Advertising vs. Sales (Prediction & SSE)

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Prediction

What are the predicted sales if the company spends €250 on advertising?

- $x = 2.5$ (since x is in hundreds of Euros)
- $\hat{y} = 1 + 2(2.5) = 1 + 5 = 6$
- **Predicted Sales:** €6000.
- *Caution: $x = 2.5$ is within the range of our data (1 to 5), so prediction is reasonable.*

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SSE Calculation

How much variability is **not** explained by the model?

$$SSE = S_{yy} - b_1 S_{xy} = 40 - (2)(20) = 40 - 40 = 0$$

Wait! $SSE = 0$? This means all data points fall exactly on the line. This is unusual in real data but happened in this simple constructed example. It implies a perfect linear fit for this sample.

Correlation - Measuring Linear Association

Purpose

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- A parameter describing the linear association in the entire population.
- Formula: $\rho = \frac{\text{Cov}(X,Y)}{\sigma_X \sigma_Y}$
- Ranges from -1 to +1. Usually unknown.

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Sample Correlation Coefficient (r)

- An estimate of ρ calculated from our sample data.
- Also called the Pearson correlation coefficient.
- Formula using sums of squares (convenient):

$$r = \frac{S_{xy}}{\sqrt{S_{xx} S_{yy}}}$$

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- **Important Cautions:**
 - Correlation only measures **linear** relationships. A strong non-linear relationship (like a parabola) could have $r \approx 0$.
 - Correlation does **not** imply causation! Just because X and Y are correlated doesn't mean X causes Y .

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- r is unitless.

Example: Advertising vs. Sales (Correlation r)

Recall our calculated values:

- $S_{xx} = 10$
- $S_{yy} = 40$
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$$r = \frac{S_{xy}}{\sqrt{S_{xx}S_{yy}}} = \frac{20}{\sqrt{(10)(40)}} = \frac{20}{\sqrt{400}} = \frac{20}{20} = 1$$

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Interpretation:

- $r = 1$ indicates a *perfect positive linear* relationship in this sample data.
- This confirms our earlier finding that all points lie exactly on the regression line ($SSE=0$).
- In real-world data, r is rarely exactly $+1$ or -1 . Values like 0.8 or -0.7 are more typical for strong relationships.

Coefficient of Determination (R^2)

Purpose

R^2 (or R-squared) provides a measure of how well the regression line explains the variation in the dependent variable (Y). It answers: "What proportion of the total variability in Y is accounted for by its linear relationship with x ?"

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Interpretation

- Range: $0 \leq R^2 \leq 1$.
- Often expressed as a percentage (multiply by 100).
- $R^2 = 0.75$ means that 75% of the observed variation in Y can be attributed to the linear model with x . The remaining 25% is unexplained (due to error ϵ).
- Higher R^2 indicates a better fit *for the sample data*.

Coefficient of Determination (R^2) - Calculation

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R^2 is the ratio of explained variability to total variability:

$$R^2 = \frac{SSR}{SST} = \frac{SST - SSE}{SST} = 1 - \frac{SSE}{SST}$$

Both methods give the same result.

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Interpretation:

- $R^2 = 1$ or 100 %.
- This means 100 % of the variability in monthly sales (in this specific sample) is explained by the linear relationship with monthly advertising spend.
- Again, this perfect fit is due to the constructed example. Real data R^2 will be less than 1.

Data Transformation - When Linearity Fails

Problem Signs

Sometimes, a straight line is not the best way to describe the relationship between X and Y . How do we know?

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Solution: Transformation

We can apply mathematical functions (like log, square root, reciprocal) to Y and/or X to:

- "Straighten out" a curved relationship.
- Stabilize non-constant variance.

Then, we perform linear regression on the *transformed* variables.

Data Transformation - Common Methods

① Logarithmic (e.g., $Y' = \ln(Y)$ or $X' = \ln(x)$):

- Useful for: Exponential growth ($Y \approx \beta_0 e^{\beta_1 x} \implies \ln(Y) \approx \ln(\beta_0) + \beta_1 x$).
- Useful for: Stabilizing variance that increases proportionally to the mean.
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4 Reciprocal (e.g., $Y' = 1/Y$ or $X' = 1/x$):

- Can sometimes linearize relationships that plateau.

Choice depends on data pattern and sometimes theoretical reasons.

New Example: Bacterial Growth

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A plot of y vs x would show a clear upward curve. **Hypothesized**

Model:** $Y \approx ae^{bx}$. **Transformation:**** Take natural log:

$\ln(Y) \approx \ln(a) + bx$. Let $Z = \ln(Y)$, $c = \ln(a)$. Fit linear model:

$Z = c + bx$.

Example: Bacterial Growth (Transformation)

Transform Y to $Z = \ln(Y)$:

x	y	$z = \ln(y)$	x^2	z^2	xz
0	1.1	0.095	0	0.009	0
1	1.9	0.642	1	0.412	0.642
2	3.5	1.253	4	1.570	2.506
3	6.5	1.872	9	3.504	5.616
4	12.0	2.485	16	6.175	9.940
$\sum x = 10$		$\sum z = 6.347$	$\sum x^2 = 30$	$\sum z^2 = 11.670$	$\sum xz = 18.704$

$$n = 5. \quad \bar{x} = 10/5 = 2. \quad \bar{z} = 6.347/5 = 1.2694.$$

Example: Bacterial Growth (Calculations on Transformed Data)

Calculate S_{xx} , S_{zz} , S_{xz} for the (x, z) data:

- $S_{xx} = \sum x^2 - \frac{(\sum x)^2}{n} = 30 - \frac{(10)^2}{5} = 30 - \frac{100}{5} = 30 - 20 = 10$

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- $S_{zz} = \sum z^2 - \frac{(\sum z)^2}{n} = 11.670 - \frac{(6.347)^2}{5} = 11.670 - \frac{40.2844}{5}$
 $= 11.670 - 8.0569 = 3.6131$

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- $S_{xz} = \sum xz - \frac{(\sum x)(\sum z)}{n} = 18.704 - \frac{(10)(6.347)}{5} = 18.704 - \frac{63.47}{5}$
 $= 18.704 - 12.694 = 6.010$

Example: Bacterial Growth (Fitted Line & Final Model)

Fit linear model $Z = c + bx$ using

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This exponential model should fit the original data better than a straight line. (R^2 on transformed data:

$$r^2 = \frac{S_{xz}^2}{S_{xx}S_{zz}} = \frac{(6.010)^2}{(10)(3.6131)} = \frac{36.1201}{36.131} \approx 0.9997).$$

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- **ALWAYS visualize data (scatter plot) and check assumptions (residual plots)!**
- Data transformations (log, sqrt, etc.) can help when the relationship isn't linear or variance isn't constant.

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Now that we can fit a line, what can we infer about the population?

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- How accurately can we predict the average response or a future individual response for a given x ?
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